Lab CA-Activity 1 (15 Marks): Agentic AI System using Modern LLM-based Tools

Activity Type - Group Activity (Max 3 students in a group)

Learning Objectives

After completing this activity, students will:

- Understand **Agentic AI beyond classical agents**
- Learn how **LLMs act as reasoning agents**
- Use **tool-using, planning, and memory-based agents**
- Compare **classical AI agents vs modern agentic systems**

Activity Description

Students will **build a small Agentic AI system** using **modern AI tools** where an LLM acts as an autonomous agent that:

- Understands a goal
- Plans steps
- Uses tools (search, calculator, documents)
- Produces an output

"This activity exposes students to modern agentic AI systems using LLM-based frameworks, which extend classical agent concepts taught in the syllabus. The focus is on tool usage, reasoning, and orchestration rather than model training, ensuring feasibility for undergraduate students."

Al Tools

Students must use **at least ONE** of the following:

Agent Frameworks

- **AutoGen**
- **CrewAI**
- **ReAct-style agents**
- **BrowserUse**
- **Openrouter (free api)**
- **Langchain**

Tools integrated with agent

- Web search APIs
- Google OAuth 2.0 (Google Calender task)
- Calculator tools
- File/document readers
- Simple databases (CSV/JSON)

- Python tools

Example Problem Statements (Students choose ONE)

1. **Research Assistant Agent**
    - Reads documents
    - Answers questions using tools
2. **Schedule meeting in google calendar**
    - Input prompt to schedule meeting
    - Automatically adds it to calendar
3. **Academic Planner Agent**
    - Plans study schedule
    - Adjusts based on constraints
4. **Healthcare Information Agent***(non-diagnostic)*
    - Collects symptoms
    - Retrieves medical guidelines
5. **Job Application Assistant Agent**
    - Analyzes resume + JD
    - Suggests improvements
6. **Travel Planning Agent**
    - Plans itinerary
    - Uses search + reasoning tools

Student Tasks

Part A – Agent Design (Modern View)

Students must explain:

- Goal of the agent
- Agent role
- Tools used
- Memory type (short-term / context)
- Planning strategy (ReAct / step-by-step)

Part B – Implementation (Modern AI Tools)

Minimum requirements:

- One **LLM-based agent**
- At least **two tools**
- Multi-step reasoning
- Observable agent behavior (logs / traces)

Example:

```
User Goal → Agent Reasoning → Tool Call → Observation → Next Action → Final Answer
```

Part C – Analysis & Comparison

Students must **compare**:

Classical Agent (Syllabus)	Modern Agent (This Activity)
Rule-based	LLM-driven
Static logic	Dynamic planning
No memory	Contextual memory
Single-step	Multi-step reasoning

Deliverables

- PDF report (5–10 pages)
- Architecture diagram
- Tool flow diagram
- Code repository / notebook
- Agent interaction logs

Evaluation Rubric (15 Marks)

Component	Marks
Use of modern tools & frameworks	4
Agent reasoning & tool orchestration	4
Architecture & design clarity	4
Analysis & reflection	3

Lab CA-Activity 2 (15 Marks): Prompt Engineering & LLM Behavior Analysis

Activity Type: **Individual activity**(or group of two only)

Learning Objectives

After completing this activity, students will be able to:

- Design **effective prompts** for LLMs
- Understand **how prompt structure affects output**
- Identify **hallucinations, bias, and failures**
- Apply **ethical considerations** in generative AI use

AI Tools

Students may use **ONLY free-access tools**, such as:

- ChatGPT (free version)

- Google Gemini (free)
- Microsoft Copilot (free)
- HuggingFace Chat models
- LM Studio / Ollama (local)

Activity Description

Students will **systematically experiment with prompt design techniques** on a chosen task and **analyze LLM behavior**.

The emphasis is on:

- Prompt **design**
- Output **comparison**
- Failure **analysis**
- Prompt **refinement**

"This activity introduces students to prompt engineering through structured experimentation using freely available generative AI tools. Students analyze LLM behaviour, identify failures and biases, and learn responsible AI usage".

Student Tasks

Step 1: Select a Task Domain (Choose ONE)

Examples:

- **Concept Tutor**
  (Explain technical concepts at different difficulty levels)
- **Resume / SOP Evaluator**
  (Provide structured feedback with constraints)
- **Code Reviewer**
  (Identify bugs + suggest improvements)
- **Healthcare Information Assistant***(non-diagnostic)*
  (Summarize guidelines, avoid medical advice)
- Educational tutoring (concept explanation)
- Resume / SOP review
- Code explanation or debugging
- Healthcare information (non-diagnostic)
- News/article summarization
- Question answering from a given text

Step 2: Design Different Prompt Types

Students must design **at least FOUR prompt types**:

1. **Zero-shot Prompt**
2. **Few-shot Prompt**
3. **Role-based Prompt**
4. **Chain-of-Thought Prompt**

The base prompt must specify:

- Role
- Output format
- Constraints
- Tone
- Safety instructions

Each prompt must be clearly written and documented.

Step 3: Execute Prompts & Capture Outputs

For each prompt:

- Prompt text
- LLM output
- Observed quality (accuracy, clarity, relevance)
- Students must run the same prompt with two different input requirements: Natural Language Output (Plain text/Paragraph) JSON Output (Structured data with specific keys) Observe if the JSON output is better or not.

Students must test the **same prompt** on **at least 5 different inputs**, such as:

- Easy vs hard inputs
- Ambiguous inputs
- Edge cases
- Incomplete information
- Contradictory information

Screenshots or logs must be included.

Step 4: Identify Prompt Failures & Biases

Students must identify **at least TWO** of the following:

- Hallucination
- Ambiguity
- Bias or ethical concern
- Over-generalization
- Incorrect assumptions

Step 5: Prompt Refinement

Students must:

- Improve the prompt using constraints, structure, or context
- Compare **before vs after** outputs
- Explain *why* the refined prompt performs better

Deliverables

- PDF report (6–8 pages)
- Prompt → Output comparison tables
- Screenshots / logs
- Reflection section (student's own learning)

Evaluation Rubric (15 Marks)

Component	Marks
Variety of prompt techniques	4
Output comparison & analysis	4
Failure/bias identification	4
Prompt refinement & reflection	3